

# Problem Statement

Evaluate the limit from the image "dailyintegral-limit-limit-1x1-2026-08-05.jpg":

$$\lim_{x \rightarrow \infty} \frac{1}{x} \sum_{n=1}^x \sin(\text{mod}(n, \pi))$$

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## Detailed Solution

To evaluate this limit, we can analyze the behavior of the sequence inside the sine function and apply Weyl's Equidistribution Theorem to convert the infinite sum limit into a definite integral.

### Step 1: Analyze the Modulo Expression

The expression  $\text{mod}(n, \pi)$  represents the remainder of  $n$  when divided by  $\pi$ . Mathematically, this can be expressed using the fractional part function:

$$\text{mod}(n, \pi) = n - \pi \left\lfloor \frac{n}{\pi} \right\rfloor = \pi \left\{ \frac{n}{\pi} \right\}$$

where  $\{y\}$  denotes the fractional part of a real number  $y$ .

### Step 2: Apply Weyl's Equidistribution Theorem

Because  $\pi$  is an irrational (and transcendental) number, its reciprocal  $1/\pi$  is also irrational. According to Weyl's Equidistribution Theorem, for any irrational number  $\alpha$ , the sequence of fractional parts  $\{n\alpha\}$  is uniformly distributed (equidistributed) in the interval  $[0, 1)$  as  $n \rightarrow \infty$ .

Substituting  $\alpha = 1/\pi$ , the sequence  $\{n/\pi\}$  is uniformly distributed in  $[0, 1)$ . Consequently, multiplying by the constant  $\pi$  implies that the sequence  $t_n = \pi\{n/\pi\} = \text{mod}(n, \pi)$  is uniformly distributed throughout the interval  $[0, \pi)$ .

### Step 3: Convert the Limit of a Sum to an Integral

For a sequence  $(t_n)$  that is equidistributed in a bounded interval  $[a, b]$ , and for any Riemann integrable function  $f$ , the limit of the arithmetic mean of the function evaluated at the sequence points equals the continuous average value of the function over that entire interval:

$$\lim_{x \rightarrow \infty} \frac{1}{x} \sum_{n=1}^x f(t_n) = \frac{1}{b-a} \int_a^b f(t) dt$$

In our specific case,  $f(t) = \sin(t)$ , the interval is  $[0, \pi]$ , and the length of the interval is  $b - a = \pi - 0 = \pi$ .

#### Step 4: Evaluate the Integral

Applying the equidistribution result, we convert the discrete limit of the sum to our continuous integral:

$$\lim_{x \rightarrow \infty} \frac{1}{x} \sum_{n=1}^x \sin(\text{mod}(n, \pi)) = \frac{1}{\pi} \int_0^\pi \sin(t) dt$$

Now, we simply evaluate the definite integral of the sine function:

$$\int_0^\pi \sin(t) dt = [-\cos(t)]_0^\pi = (-\cos(\pi)) - (-\cos(0))$$

Since  $\cos(\pi) = -1$  and  $\cos(0) = 1$ :

$$(-(-1)) - (-1) = 1 + 1 = 2$$

Therefore, dividing by the interval length  $\pi$ , we find the limit:

$$\frac{1}{\pi}(2) = \frac{2}{\pi}$$

**Final Answer:**

$$\frac{2}{\pi}$$